S Mukhesh Vital Babu

Amaravathi, India

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Education

VIT-AP University — Sep. 2022 – May 2026
B.Tech in Computer Science with Cyber Security — *Amaravathi, Andhra Pradesh*

NRI Junior College — July 2020 – August 2022
Intermediate — *Guntur, Andhra Pradesh*

Gheetam School — May 2020
Class 10 — *Guntur, Andhra Pradesh*

Skills

Programming & Data: Python (NumPy, Pandas, Scikit-learn), SQL, MATLAB, Java
Machine Learning & AI: Supervised & Unsupervised Learning, CNNs, Model Training & Fine-tuning, Feature Engineering, Cross-validation, Evaluation Metrics, Model Optimization, Exploratory Data Analysis (EDA)
Generative AI: Large Language Models (LLMs), Prompt Engineering, Semantic Retrieval, Transformer-based Models
Frameworks & Tools: PyTorch, ONNX, LangChain, Hugging Face, OpenCV, FastAPI, Docker, Git, GitHub, VS Code
Databases & Systems: SQL, MongoDB, ChromaDB, REST APIs, Linux (Basic)

Coursework

Machine Learning, Data Structures Algorithms, Database Management Systems, Statistical Methods, Software Engineering, MATLAB Programming

Projects

ResearchPilot AI – Autonomous Multi-Agent Research Assistant | *Python, LangGraph, Google Gemini, Tavily API*

- Developed an autonomous multi-agent research assistant that transforms user queries into structured, source-backed research reports.
- Implemented a five-agent LangGraph workflow comprising Planner, Web Search, Synthesizer, Critique, and Report Writer agents.
- Integrated Google Gemini and Tavily Search API for intelligent research planning, real-time web search, information synthesis, and citation generation.
- Built an interactive Streamlit dashboard with research configuration, live execution progress, report visualization, and Markdown export.

Asthma Diagnosis: A Machine Learning Perspective | *Python, MATLAB*

- Conducted a machine learning-based study on asthma severity prediction using structured clinical datasets.
- Performed exploratory data analysis (EDA) and evaluated models using statistical metrics and cross-validation.
- Analyzed model generalization, validation stability, and performance trade-offs.
- Achieved **97.8% accuracy**, demonstrating strong predictive performance and reliable analytical insights.

Pharmaceutical Drug Matching (Generative AI) | *Python, LangChain, Hugging Face, ChromaDB, spaCy*

- Built a semantic retrieval and matching system for pharmaceutical drug data using vector embeddings and metadata-aware indexing.
- Designed modular retrieval logic combining vector search, structured validation, and rule-based filtering pipelines.
- Implemented prompt-driven reasoning, summarization, and entity extraction using transformer-based language models.
- Benchmarked retrieval accuracy and response quality across multiple configurations to improve consistency and reliability.
- Evaluated model degradation across data variations and retrained models to maintain accuracy and robustness.

AI-Enhanced Online Auction Platform | *React, Node.js, Express.js, MongoDB, JWT, WebSockets*

- Developed a full-stack online auction platform supporting real-time bidding with persistent user and session state.
- Implemented WebSocket-based live bid updates with concurrency handling, reconnection safety, and event consistency.
- Designed secure authentication and role-based access control using JWT with modular REST APIs.

Certifications

- Cyber Security Bootcamp – Let's Upgrade
- Oracle Generative AI Professional
- Cyber Security Essentials – Cisco Systems
- AWS Cloud Foundations – AWS Academy Graduate
- AWS Cloud Architecture – AWS Academy Graduate

Leadership & Activities

Competitive Programming Club — Member

Outreach Club — Active Member

Interests

Capture the Flag (CTF) Challenges, Research, Music